

Region Resonances

Data Appendix

Source Data, Tables, and Reference Materials

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Companion documents: Summary · Technical Appendix · Methods Primer · Region Reference — Identity

D.1 Corpus

The Region Resonances corpus consists of 59 region passages, one per wine region. Each passage is the concatenation of the region’s one-word identity metaphor with its full Layer 1 identity narrative. Passage lengths range from 280 to 320 words. The metaphors and the source narratives are produced and maintained by the Soul of Wine project; the full source narratives are available in the companion document *Region Reference — Identity*, which is shared between Region Resonances and Region Affinities.

The 59 regions and their identity metaphors, sorted alphabetically:

Region	Country	Identity Metaphor
Alsace	France	<i>Duality</i>
Alto Adige	Italy	<i>Precision</i>
Baden	Germany	<i>Warmth</i>
Barossa Valley	Australia	<i>Fortitude</i>
Beaujolais	France	<i>Joy</i>
Bordeaux	France	<i>Business</i>
Burgundy	France	<i>Devotion</i>
Campania	Italy	<i>Memory</i>
Catalonia	Spain	<i>Identity</i>
Central Otago	New Zealand	<i>Adventure</i>
Champagne	France	<i>Society</i>
Châteauneuf-du-Pape	France	<i>Family</i>
Columbia Valley	USA	<i>Determination</i>
Dalmatian Coast	Croatia	<i>Tranquility</i>
Douro	Portugal	<i>Endurance</i>
Etna	Italy	<i>Awakening</i>
Finger Lakes	USA	<i>Conviction</i>
Friuli-Venezia Giulia	Italy	<i>Dialogue</i>
Galicia	Spain	<i>Longing (Morriña)</i>
Goriška Brda	Slovenia	<i>Fortune</i>
Hawke’s Bay	New Zealand	<i>Confidence</i>
Hunter Valley	Australia	<i>Defiance</i>
Jura	France	<i>Eccentricity</i>
Kamptal	Austria	<i>Discipline</i>
Liguria	Italy	<i>Intimacy</i>
Loire	France	<i>Sentimentality</i>
Macedonia	Greece	<i>Austerity</i>
Maipo Valley	Chile	<i>Pride</i>
Margaret River	Australia	<i>Composure</i>

Region	Country	Identity Metaphor
Marlborough	New Zealand	<i>Assertion</i>
Mendoza	Argentina	<i>Reinvention</i>
Mosel	Germany	<i>Poetry</i>
Nahe	Germany	<i>Subtlety</i>
Napa Valley	USA	<i>Ambition</i>
Northern Rhône	France	<i>Solitude</i>
Paso Robles	USA	<i>Independence</i>
Patagonia	Argentina	<i>Extremity</i>
Pfalz	Germany	<i>Generosity</i>
Piedmont	Italy	<i>Philosophy</i>
Provence	France	<i>Pleasure</i>
Rheingau	Germany	<i>Nobility</i>
Ribera del Duero	Spain	<i>Severity</i>
Rioja	Spain	<i>Patience</i>
Santa Barbara	USA	<i>Serendipity</i>
Santa Cruz Mountains	USA	<i>Obsession</i>
Santorini	Greece	<i>Survival</i>
Sardinia	Italy	<i>Stubbornness</i>
Sicily	Italy	<i>Resurrection</i>
Sonoma	USA	<i>Authenticity</i>
Steiermark	Austria	<i>Clarity</i>
Stellenbosch	S. Africa	<i>Aspiration</i>
Swartland	S. Africa	<i>Rebellion</i>
Tokaj	Hungary	<i>Melancholy</i>
Tuscany	Italy	<i>Art</i>
Veneto	Italy	<i>Commerce</i>
Wachau	Austria	<i>Monumentality</i>
Wagram	Austria	<i>Earth</i>
Walla Walla	USA	<i>Community</i>
Willamette Valley	USA	<i>Idealism</i>

D.2 Embedding Model

The embedding map is OpenAI’s `text-embedding-3-small` model. Key parameters:

Parameter	Value
Model name	<code>text-embedding-3-small</code>
Output dimension	$d = 1536$
Output normalisation	Unit-norm ($\ \mathbf{v}\ _2 = 1$)
Input format	UTF-8 plain text, each passage as “ <code>{metaphor}. {narrative}</code> ”
Tokeniser	cl100k_base (OpenAI standard)
Max input tokens	8,191
Provider	OpenAI Embeddings API

The model is treated throughout as a fixed, opaque function. Region Resonances does not fine-tune, retrain, or adjust the model in any way. If the model is updated by the provider, the corpus embeddings are regenerated and the change is documented.

D.3 Display Calibration Constants

The clip-and-rescale transformation that converts raw cosine similarity values to displayed scores in $[0, 100]$ uses two calibration constants:

Constant	Value
$s_{\min}$ (lower clip threshold)	0.15
$s_{\max}$ (upper clip threshold)	0.70
Display range	$[0, 100]$

These constants were calibrated against a 60-query test set described in Section D.4. They are not learned during runtime; they are fixed parameters of the deployed system.

D.4 Calibration Test Set

A test set of 60 English-language queries was assembled during development to characterise the empirical operating range of the matcher against the 59-region corpus. The set spans three categories:

- *Single-word queries* (24 items). Examples: *patience*, *rebellion*, *melancholy*, *ambition*, *precision*, *warmth*.
- *Short phrase queries* (24 items). Examples: *old soul*, *quiet confidence*, *against the odds*, *constructed identity*, *slow time*.
- *Longer fragment queries* (12 items). Examples: *the feeling of late summer*, *a culture that does not need external confirmation*, *regions where the past is more vivid than the present*.

The test set is not used to train or tune the embedding model. It is used solely to (a) confirm the empirical concentration of cosine similarities in $[0.15, 0.70]$, (b) calibrate the display constants $s_{\min}$ and $s_{\max}$, and (c) provide the operating-characteristics statistics reported in the Summary.

D.5 Empirical Operating Characteristics

Distributional statistics observed across the calibration test set:

Statistic	Value
Minimum observed cosine similarity	0.13
Maximum observed cosine similarity	0.71
Median top-match (rank 1) cosine	0.52
Median fifth-match (rank 5) cosine	0.43
Median top-match displayed score	≈ 67
Median fifth-match displayed score	≈ 51
Median top-1 to top-5 cosine spread	0.09
Fraction of queries with at least one match ≥ 0.50	87%

These statistics describe the distribution observed on the calibration set against the 59-region corpus and the deployed embedding model. They are not asymptotic guarantees.

D.6 Pipeline Parameters

A consolidated reference of all parameters used in the deployed system.

Parameter	Value
Corpus size n	59 regions
Passage construction	"{metaphor}. {narrative}"
Embedding model	text-embedding-3-small
Embedding dimension d	1536
Output normalisation	Unit-norm
Similarity metric	Cosine (equivalent to inner product on unit sphere)
Top- k display	5
Display transformation	Clip-and-rescale, linear in cosine
$s_{\min}, s_{\max}$	0.15, 0.70
Display range	[0, 100]
Determinism	Fully deterministic given corpus and embedding model

D.7 Source Data Files

The Region Resonances tool is shipped as a single-page React application. Static assets in the `public/` directory:

File	Contents
<code>region_embeddings.json</code>	59 region records, each containing region name, identity metaphor, source passage text, and the 1536-dimensional embedding vector. Embedding metadata block specifies the model, dimension, and generation timestamp.
<code>layer1-narratives.json</code>	59 region narratives in their full Layer 1 form, used for the on-page narrative display when a region is selected.

The corpus embeddings are regenerated from the source narratives by an offline Python script that calls the OpenAI Embeddings API; the script is part of the project repository.

D.8 Reproducibility

The matching procedure is fully deterministic given the corpus and the embedding model. Identical queries produce identical rankings. Software versions: Python 3.11 (offline pre-computation), Node.js 20 (web tool runtime), OpenAI Python SDK 1.30.